

Quant Research Scout

Daily report - 2026-05-11

7 findings saved this run

Digest

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****Session note:**** All academic paper APIs (arxiv, Semantic Scholar, SSRN, NBER, Reddit, HN, Quantocracy) returned 403 from the proxy allowlist; a prior automated run earlier today captured the arxiv findings. This session extended coverage via GitHub topic scraping (regime-detection, volatility-forecasting). Total today: 7 findings across 2 sources.

Top Findings (Ranked)

1. Adaptive VaR Control for Option Books via Sequential Conformal Recalibration

****Source:**** arxiv | ****Relevance:**** 0.85 | ****Replicability:**** 3/5

****URL:**** <https://arxiv.org/abs/2604.03499>

Develops a tail-risk framework for standardized option books combining a base conditional-quantile forecast with sequential conformal recalibration. Tested on SPX options 2018-2025; uncalibrated forecasts systematically underestimate downside risk, and sequential recalibration restores coverage and improves rolling-window tail stability. Tackles the "next-day marking friction" problem - what to do when same-contract quotes are unavailable for marking.

****Why it matters:**** Conformal prediction applied to option-book VaR is practical and distribution-free. Sequential recalibration is a tractable solution to a real production problem (stale quotes at EOD). Direct plug-in for the risk engine's options module.

2. Minimum Regime Performance (MRP): Measuring Strategy-Decay Risk

****Source:**** arxiv | ****Relevance:**** 0.83 | ****Replicability:**** 4/5

****URL:**** <https://arxiv.org/abs/2604.08356> | ****Code:**** github.com/nolanalexander/minimum_regime_performance

Introduces MRP - the lowest risk-adjusted return realized by a strategy **across distinct historical regimes** - as a lower bound on durability. Applied to a broad universe of factor strategies, exposing an efficiency-vs-resilience trade-off: high long-term Sharpe does not imply high MRP. Code is open.

****Why it matters:**** MRP turns "strategy decay" from anecdote into a regime-conditional measurable. Directly composable with the regime classifier layer of the quantamental engine - use detected regimes to compute live MRP for active strategies. The code is open and the methodology is clean.

3. QuantRiskEngine: C++/Python Risk Engine with GARCH-LSTM and VaR Backtesting

****Source:**** github | ****Relevance:**** 0.83 | ****Replicability:**** 4/5****URL:**** <https://github.com/BlazejRadzik/QuantRiskEngine>

A portfolio risk management platform with a C++ SIMD-accelerated Monte Carlo core (~9x speedup over NumPy), hybrid GARCH-LSTM vol modeling, VaR/CVaR estimation, and formal backtesting via Kupiec POF and Christoffersen independence tests. Includes risk_report.pdf output.

****Why it matters:**** Architecture directly mirrors the user's project. The SIMD Monte Carlo + GARCH-LSTM pairing and the inclusion of Kupiec/Christoffersen backtests as first-class outputs are both reference-worthy design decisions. Mine for the backtesting module structure and the C++/Python bridge pattern.

4. Skewness Dispersion as a Market Return Predictor

****Source:**** arxiv | ****Relevance:**** 0.80 | ****Replicability:**** 3/5****URL:**** <https://arxiv.org/abs/2604.07870>

Cross-sectional dispersion of firm-level realized skewness predicts aggregate market returns - significantly, negatively, and incrementally to a 50-predictor benchmark including VRP, term spread, and P/E. Only 6 of 50 established predictors beat its OOS bar.

****Why it matters:**** Skewness dispersion is computable from options data or from daily returns, making it a candidate factor to slot into the regime-conditional risk overlay. The OOS evidence is meaningful given the large comparison set.

5. HARNet: Neural Extension of HAR Realized Volatility Model

****Source:**** github | ****Relevance:**** 0.78 | ****Replicability:**** 5/5****URL:**** <https://github.com/mdsunivie/HARNet> | ****Paper:**** arXiv:2205.07719

TensorFlow implementation of HARNet, which replaces the HAR model's linear regression with a learned nonlinear mapping while preserving the daily/weekly/monthly lag structure. Backed by a published arxiv paper, 29 stars.

****Why it matters:**** HAR is the benchmark to beat for RV forecasting. HARNet is a principled, paper-backed upgrade with open code - the right starting point for the vol-forecasting module. Convergence with the Options-driven RV paper (#6 below), which explicitly evaluates HAR augmentations.

Notable-But-Borderline

****lob-regime-scanner**** (github.com/CameronScarpati/lob-regime-scanner, already in store): Reported Sharpe of 1.8-2.5 on a walk-forward test should be treated skeptically - no third-party replication and only 3 stars. But the feature set (multi-level OFI, VPIN, Kyle's lambda) and the C++/pybind11 LOB architecture are reference-quality for real-time microstructure feature extraction. Mine the feature definitions, not the headline numbers.

****Options-driven RV forecasting via rough vol**** (arxiv:2604.02743, in store, rel=0.72): Extracts spot vol from options under a rough SV model and feeds it into HAR. The deep-learning surrogate for large-panel estimation is a notable side contribution. Low replicability (2/5) because the rough-vol extraction step requires calibration infrastructure not included.

Multi-Source Convergence

****Vol-forecasting thread is hot today:**** HARNet (GitHub), QuantRiskEngine (GitHub), Options-driven RV paper (arxiv), and Adaptive VaR paper (arxiv) all touch volatility forecasting from different angles. The Options-RV paper explicitly augments HAR - directly connecting to HARNet. This four-source convergence on vol-forecasting warrants a dedicated investigation session.

****Regime -> Strategy durability link:**** MRP (arxiv) connects cleanly to the lob-regime-scanner and the Regime Performance paper to form a two-layer stack: (1) detect current regime, (2) evaluate whether active strategies are known to degrade in that regime via MRP.

Next Investigation Suggestion

****Deep-dive on conformal prediction for real-time risk.**** The Adaptive VaR paper (finding #1) applies sequential conformal recalibration to option-book VaR. The broader conformal prediction literature has extended this to time-series forecasting with online coverage guarantees - which is exactly the constraint a real-time risk engine wants. Investigate whether `MAPIE` (Python conformal prediction library) or `nonconformist` can wrap existing GARCH/LSTM forecasters to give distribution-free VaR intervals, and whether the sequential (online) variant maintains coverage under regime shifts.

7 findings saved today. Sources: arxiv (4), github (3). All academic APIs blocked by proxy allowlist during this session - prior automated run captured arxiv findings.

1. Adaptive VaR Control for Standardized Option Books under Marking Frictions

<https://arxiv.org/abs/2604.0003>

Source: arxiv Type: model Relevance: 0.85 Replicability: ***-- (3/5)

Authors: Tengan Zhong

Published: 2026-04-03

TLDR

(not provided)

Summary

Develops a tail-risk framework for standardized option books that combines a base conditional-quantile forecast with sequential conformal recalibration. Tested on SPX option data 2018-2025; the uncalibrated base systematically underestimates downside risk, and sequential recalibration restores coverage and improves rolling-window tail stability. Provides an approximate one-step exceedance-control result and quantifies the error from next-day marking when same-contract quotes are unavailable.

Key insight

Sequential conformal recalibration on top of a base conditional-quantile forecast is a tractable way to keep VaR coverage honest on option-book tails through regime shifts, including the next-day marking-friction problem when same-contract quotes vanish.

Plain English (ELI5)

(not provided)

Tags: var, options, conformal, tail-risk, spx

2. Measuring Strategy-Decay Risk: Minimum Regime Performance and the Durability of Systematic Investing

<https://arxiv.org/abs/2026.04090>

Source: arxiv Type: regime_classifier Relevance: 0.83 Replicability: ****- (4/5)

Authors: Nolan Alexander, Frank Fabozzi

Published: 2026-04-09

TLDR

(not provided)

Summary

Introduces Minimum Regime Performance (MRP) - the lowest realized risk-adjusted return across distinct historical regimes - as a lower bound on a systematic strategy's durability. Applied to a broad universe of factor strategies, MRP exposes an efficiency-vs-resilience trade-off: high long-term Sharpe does not imply high MRP. Code on GitHub at github.com/nolanalexander/minimum_regime_performance.

Key insight

MRP turns 'strategy decay' from anecdote into a regime-conditional measurable that complements market and liquidity risk and helps rank factor strategies by robustness rather than by mean Sharpe.

Plain English (ELI5)

(not provided)

Tags: regime, factor-decay, systematic, fabozzi, code

3. Skewness Dispersion and Stock Market Returns

<https://arxiv.org/abs/2026.04091>

Source: arxiv Type: model Relevance: 0.80 Replicability: ***-- (3/5)

Authors: Mykola Babiak, Jozef Barunik, Josef Kurka

Published: 2026-04-09

TLDR

(not provided)

Summary

Cross-sectional dispersion in firm-level realized skewness is significantly and negatively related to future stock market returns. The signal's predictive content is incremental to a comprehensive set of 50 established predictors in both in-sample and out-of-sample tests; only six other variables beat that bar.

Key insight

Cross-sectional skewness dispersion is a previously-overlooked aggregate market-return predictor that survives a horse race against most of the predictor zoo, including in OOS tests.

Plain English (ELI5)

(not provided)

Tags: factor, skewness, predictability, barunik

4. lob-regime-scanner: HMM-based market microstructure regime detection for cryptocurrency order books

<https://github.com>

Source: github Type: regime_classifier Relevance: 0.78 Replicability: ***** (5/5)

Authors: Cameron Scarpati

Published: 2026-04

TLDR

(not provided)

Summary

End-to-end open-source pipeline that ingests Tardis L2 snapshots, computes 30+ microstructure features (multi-level OFI, VPIN, Kyle's lambda, spread/vol/autocorr), fits a 3-state Gaussian HMM (selected by BIC across $K=2..5$), decodes regimes via Viterbi, and visualizes everything in a 4-panel Plotly Dash dashboard. C++/pybind11 LOB engine handles 1M+ updates/sec; 158-test pytest suite. Reports a regime-conditional strategy with claimed Sharpe 1.8-2.5 (test set, walk-forward).

Key insight

Demonstrates a fully reproducible HMM-on-microstructure regime classifier with a working dashboard architecture - the kind of reference design to mine for OFI/VPIN feature definitions and a real-time regime overlay pattern, even if the headline Sharpe is treated skeptically (only 3 stars, no out-of-sample period beyond train/test split).

Plain English (ELI5)

(not provided)

Tags: hmm, microstructure, ofi, vpin, lob, dashboard, open-source

5. On options-driven realized volatility forecasting: Information gains via rough volatility model

<https://arxiv.org/abs/2508.10000>

Source: arxiv Type: model Relevance: 0.72 Replicability: **--- (2/5)

Authors: Zheqi Fan, Meng (Melody) Wang, Yifan Ye

Published: 2026-04-03

TLDR

(not provided)

Summary

Extracts model-based spot volatility from traded options under a rough stochastic volatility model (iterative two-step estimation a la Andersen et al. 2015a) and feeds it into the HAR family. Benchmarked against Heston, Bates, SVCJ and the VIX, the augmented HAR-RV-RHeston improves daily realized-volatility forecasting and sustains it. A deep-learning surrogate accelerates large-panel estimation.

Key insight

Spot volatility extracted from options under a rough-vol model carries forecast information that the VIX (and standard SV models) miss - a candidate input feature for HAR-style RV forecasters in a quantamental risk engine.

Plain English (ELI5)

(not provided)

Tags: volatility, options, rough-vol, har, deep-learning

6. QuantRiskEngine: C++/Python Portfolio Risk Engine with GARCH-LSTM and Backtesting

<https://github.com>

Source: github Type: model Relevance: 0.83 Replicability: ****- (4/5)

Authors: BlazejRadzik

TLDR

A hybrid C++/Python portfolio risk engine that combines fast Monte Carlo simulation with GARCH-LSTM volatility and includes proper statistical backtesting of VaR accuracy.

Summary

A portfolio risk management platform combining a C++ Monte Carlo engine (using SIMD kernels for ~9x speedup over NumPy) with hybrid GARCH-LSTM volatility modeling, VaR/CVaR estimation, and formal backtesting via Kupiec POF and Christoffersen independence tests.

Key insight

Pairing a compiled C++ Monte Carlo core (SIMD-accelerated) with an LSTM-augmented GARCH vol model yields production-quality VaR estimates fast enough for real-time risk pipelines, and the Kupiec/Christoffersen test suite provides formal statistical validation of coverage.

Plain English (ELI5)

Value-at-Risk tells you how much money a portfolio could lose on a bad day. Getting this right requires both a good volatility estimate and a fast simulation. This project glues together a C++ simulation engine (very fast because it uses CPU tricks) with a GARCH-LSTM model that blends classical volatility forecasting with neural networks. It also checks whether the VaR estimates are actually accurate using two standard statistical tests (Kupiec and Christoffersen). The architecture closely mirrors what the user is building.

Tags: var, cvar, monte-carlo, garch, lstm, c++, simd, kupiec, christoffersen, backtesting, vol-forecasting, factors

7. HARNet: Neural Extension of the HAR Model for Realized Volatility Forecasting

<https://github.com>

Source: github Type: model Relevance: 0.78 Replicability: ***** (5/5)

Authors: mdsunivie

TLDR

Open TensorFlow implementation of HARNet, a neural network that upgrades the classic HAR realized-volatility model with learned nonlinear structure while keeping its multi-timescale design.

Summary

TensorFlow implementation of HARNet, a neural network architecture that generalizes the Heterogeneous Autoregressive (HAR) model for realized volatility forecasting. HAR is the workhorse model for RV prediction; HARNet replaces its linear structure with a learned nonlinear mapping while preserving the multi-scale (daily, weekly, monthly) lag structure that makes HAR work. Accompanied by the arxiv paper arXiv:2205.07719.

Key insight

The HAR model beats most complex ML models on realized vol because its lag structure matches volatility clustering timescales; HARNet keeps that structure but learns nonlinear interactions, offering a principled upgrade path with open code and a published paper.

Plain English (ELI5)

Realized volatility - how much a market actually moved today - can be predicted surprisingly well by just looking at how much it moved over the past day, week, and month. The HAR model has been doing this with a simple linear formula for years. HARNet swaps the linear formula for a small neural network but keeps the same three-timescale structure, so it learns richer patterns without throwing away what made HAR good. The code is open, the paper is on arxiv, and the whole thing runs in TensorFlow - making it straightforward to drop into a vol forecasting pipeline.

Tags: har, realized-volatility, tensorflow, neural-network, arxiv:2205.07719, vol-forecasting, ml